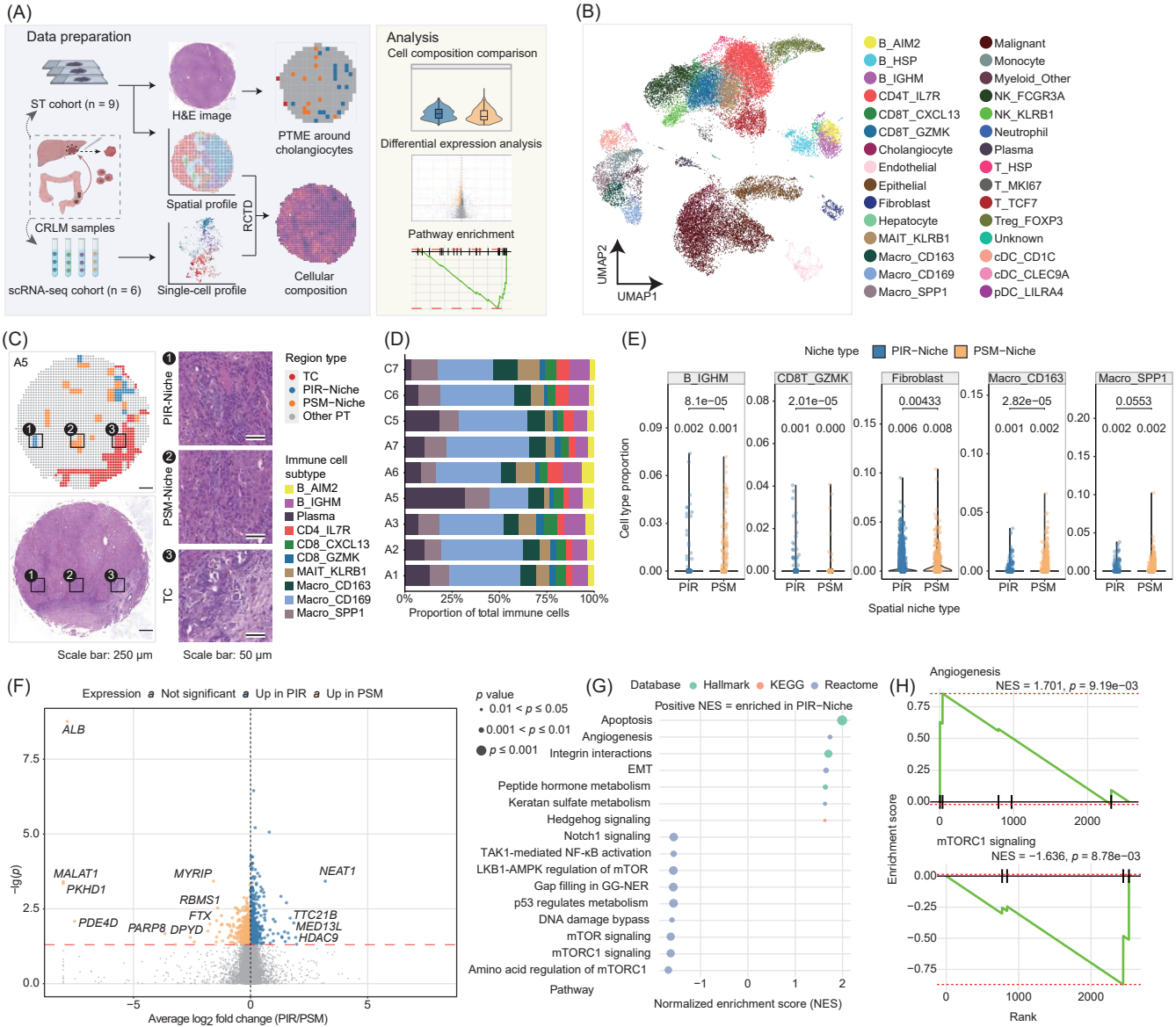

(A)
Data preparation
ST cohort (n = 9)
CRLM samples
scRNA-seq cohort (n = 6)
H&E image
Spatial profile
Single-cell profile
RCTD
PTME around cholangiocytes
Cellular composition
Analysis
Cell composition comparison
Differential expression analysis
Pathway enrichment
(B)
UMAP2
UMAP1
B_AIM2
B_HSP
B_IGHM
CD4T_IL7R
CD8T_CXCL13
CD8T_GZMK
Cholangiocyte
Endothelial
Epithelial
Fibroblast
Hepatocyte
MAIT_KLRB1
Macro_CD163
Macro_CD169
Macro_SPP1
Malignant
Monocyte
Myeloid_Other
NK_FCGR3A
NK_KLRB1
Neutrophil
Plasma
T_HSP
T_MKI67
T_TCF7
Treg_FOXP3
Unknown
cDC_CD1C
cDC_CLEC9A
pDC_LILRA4
(C)
A5
PIR-Niche
PSM-Niche
TC
Region type
TC
PIR−Niche
PSM−Niche
Other PT
Immune cell subtype
B_AIM2
B_IGHM
Plasma
CD4_IL7R
CD8_CXCL13
CD8_GZMK
MAIT_KLRB1
Macro_CD163
Macro_CD169
Macro_SPP1
Scale bar: 250 μm
Scale bar: 50 μm
(D)
C7
C6
C5
A7
A6
A5
A3
A2
A1
0%
25%
50%
75%
100%
Proportion of total immune cells
(E)
Niche type
PIR−Niche
PSM−Niche
B_IGHM
8.1e−05
0.002
0.001
CD8T_GZMK
2.01e−05
0.001
0.000
Fibroblast
0.00433
0.006
0.008
Macro_CD163
2.82e−05
0.001
0.002
Macro_SPP1
0.0553
0.002
0.002
Cell type proportion
PIR
PSM
Spatial niche type
(F)
Expression a Not significant a Up in PIR a Up in PSM
ALB
MALAT1
PKHD1
MYRIP
NEAT1
RBMS1
FTX
PDE4D
PARP8
DPYD
TTC21B
MED13L
HDAC9
−lg(p)
Average log2 fold change (PIR/PSM)
(G)
p value
0.01 < p ≤ 0.05
0.001 < p ≤ 0.01
p ≤ 0.001
Database
Hallmark
KEGG
Reactome
Positive NES = enriched in PIR−Niche
Apoptosis
Angiogenesis
Integrin interactions
EMT
Peptide hormone metabolism
Keratan sulfate metabolism
Hedgehog signaling
Notch1 signaling
TAK1-mediated NF-κB activation
LKB1-AMPK regulation of mTOR
Gap filling in GG-NER
p53 regulates metabolism
DNA damage bypass
mTOR signaling
mTORC1 signaling
Amino acid regulation of mTORC1
Pathway
Normalized enrichment score (NES)
(H)
Angiogenesis
NES = 1.701, p = 9.19e−03
mTORC1 signaling
NES = −1.636, p = 8.78e−03
Enrichment score
Rank